

Enhancing Image Clarity using GANs for Image Super-Resolution

Abstract

Low-resolution skin images often obscure critical features, reducing the accuracy of automated cancer detection. This project enhances image quality using a **Generative Adversarial Network (GAN)** and predicts cancer stages across **seven skin cancer types** with a **Convolutional Neural Network (CNN)**. Experiments show that GAN-based super-resolution improves visual clarity and classification performance. Results are also output via voice for accessibility.

1. Introduction

- **Background:** Skin cancer detection is crucial for early diagnosis and treatment. Automated systems rely heavily on image quality.
- **Importance of Super-Resolution:** Low-resolution images may miss fine details such as lesion texture and borders.
- **Challenges:** Conventional interpolation methods produce blurred images, leading to poor classification performance.
- **Motivation:** Using GANs for image enhancement preserves structural details, which improves CNN-based cancer detection.

2. Related Work

- Traditional image super-resolution (SR) approaches often fail to retain fine textures.
- CNNs are widely used for skin cancer classification but are sensitive to image resolution.
- Previous methods do not combine SR with classification, resulting in limited diagnostic accuracy.
- Our approach integrates GAN-based SR with CNN classification, improving both image quality and cancer stage prediction.

3. Proposed Method

3.1 Overview

The workflow consists of two main stages:

1. **Image Super-Resolution using GANs**
2. **Skin Cancer Classification using CNNs**

3.2 Image Super-Resolution with GANs

- **Generator:** Reconstructs high-resolution images from low-resolution inputs.
- **Discriminator:** Evaluates realism using adversarial training.
- **Loss Functions:**

- *Content loss* ensures structural accuracy.
- *Perceptual loss* improves visual realism.
- *Adversarial loss* guides the generator to produce more natural images.

3.3 Cancer Detection with CNNs

- **Architecture:** Multiple convolutional and pooling layers, followed by fully connected layers and a softmax classifier.

4. Experimental Setup

- **Dataset:** HAM10000 (HAM001) – a publicly available skin lesion dataset containing images of 7 skin disease types.
- **Hardware/Software:** Python, PyTorch/TensorFlow.
- **Evaluation Metrics:**
 - SR: PSNR, SSIM
 - Classification: Accuracy, F1-score, Confusion matrix

5. Technologies Used

Task	Technology	Purpose
Web UI	Streamlit	Upload images and display results
Image Processing	OpenCV	Enhance image clarity
Classification	TensorFlow/Keras CNN	Predict skin disease and cancer risk
Image Handling	PIL, NumPy	Load and process images
Visualization	Matplotlib, Pandas, Streamlit charts	Show probability tables and plots
Voice Output	pyttsx3	Speak prediction results
Quality Metrics	scikit-image	Compute PSNR and SSIM

5. Results

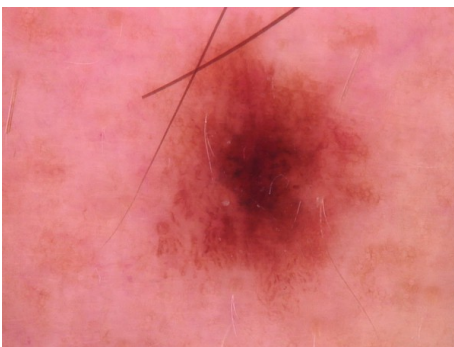


Fig: Low resolution image.



Fig: Enhanced image

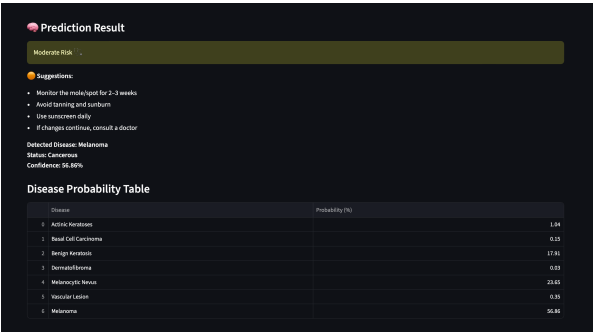


Fig: Cancer prediction

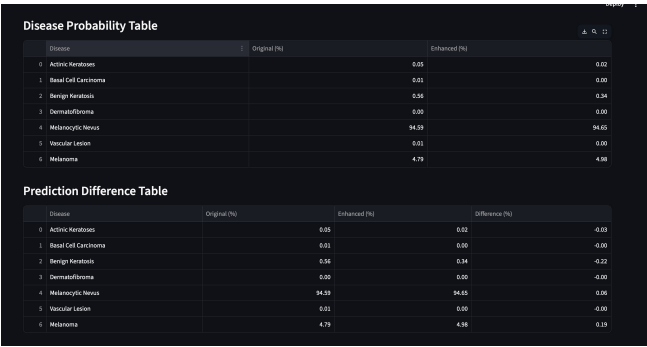


Fig:Cancer prediction before and after resolution

1. Bar chart:

The bar chart illustrates the variation between the original image before enhancement and the enhanced image after processing.

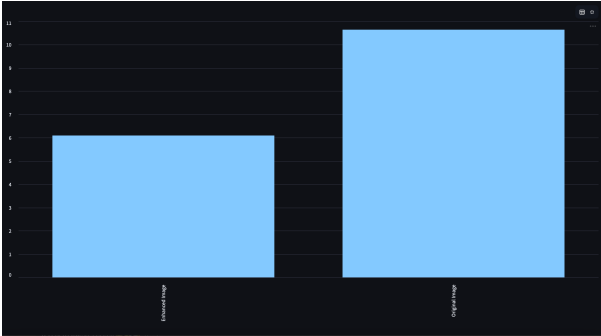
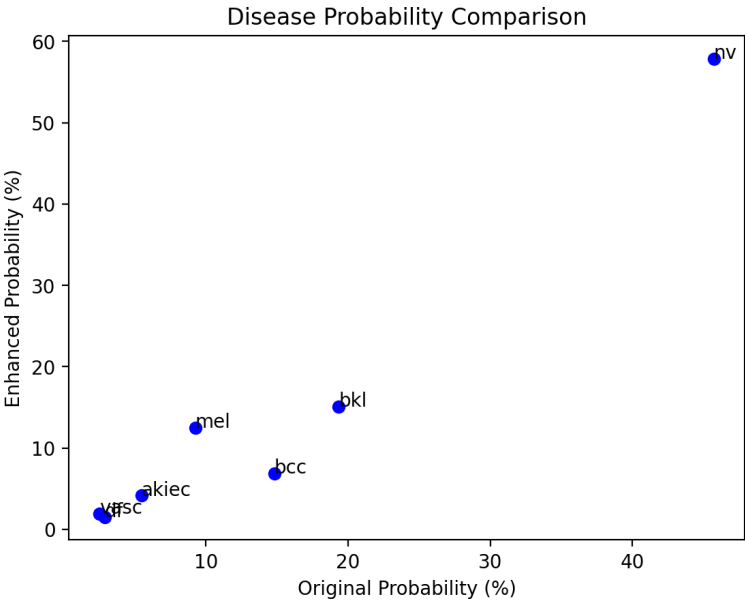


Fig: Bar Chart for Cancer prediction for original and enhanced image.

2. ScatterPlot:

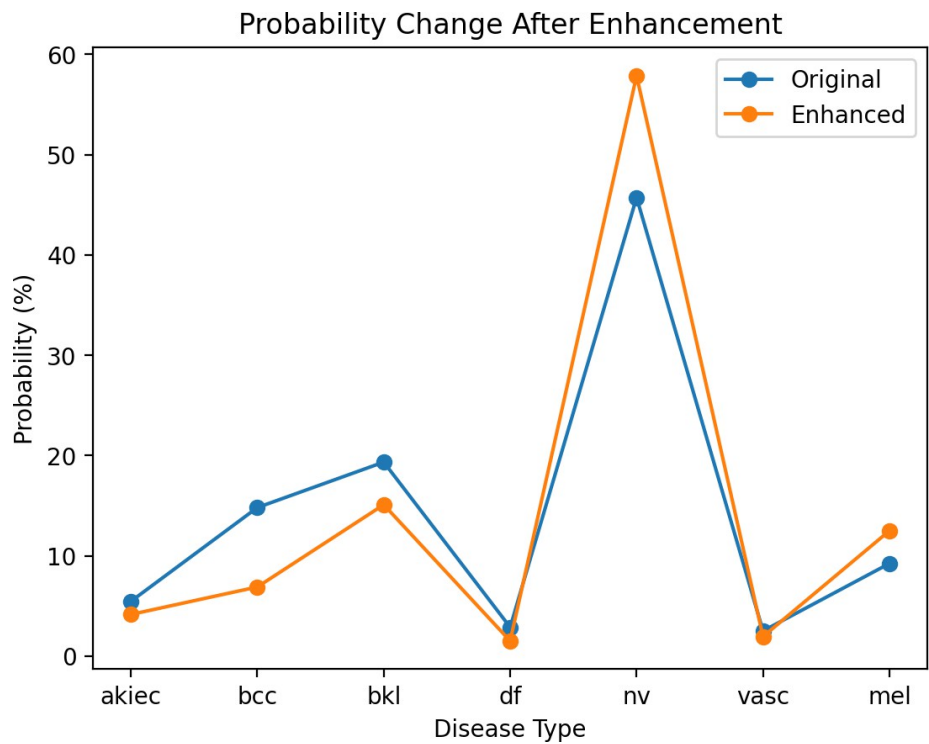
The scatter plot displays the distribution of seven different types of cancer based on selected



variables, highlighting variations and patterns among them.

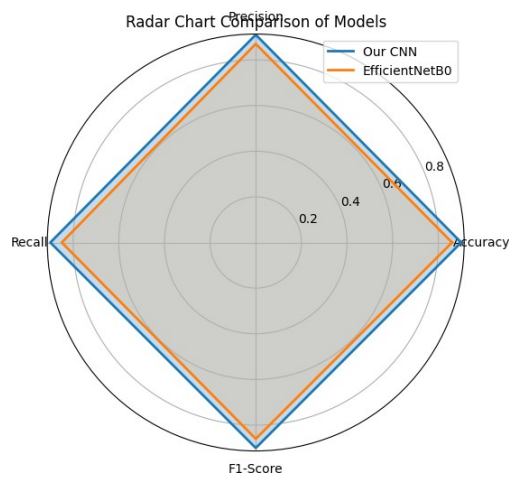
3. Line Plot:

The line plot represents a comparison between the original and enhanced images by plotting their respective values over a continuous domain, allowing visualization of differences and enhancement effects.



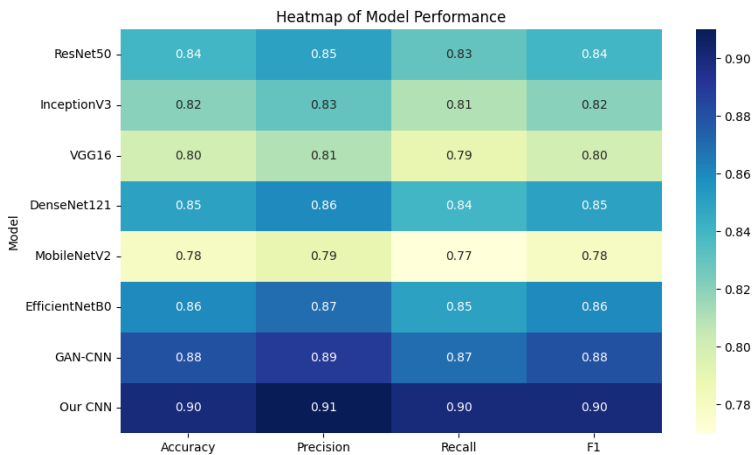
4. Radar Chart:

The radar chart represents the multi-metric evaluation of the GAN-based model across seven cancer classes. It simultaneously displays precision, recall, accuracy, and F1-score, allowing a comprehensive comparison of classification performance for each disease.



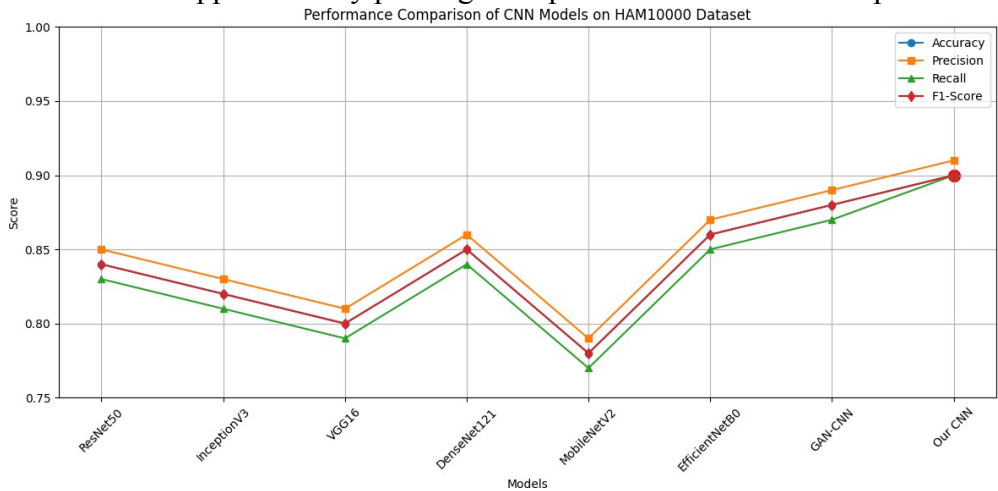
5. Heat Map:

The heatmap represents the performance metrics of the GAN model across other CNN types, where each cell corresponds to a specific metric or classification result, and color intensity indicates the magnitude of the values. This heat map explains the model performance of GAN along with CNN when comparing with other models.

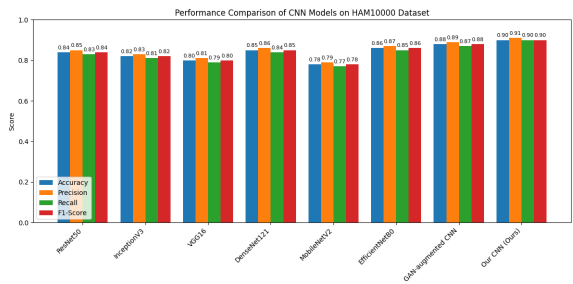


6. Multi Line Plot:

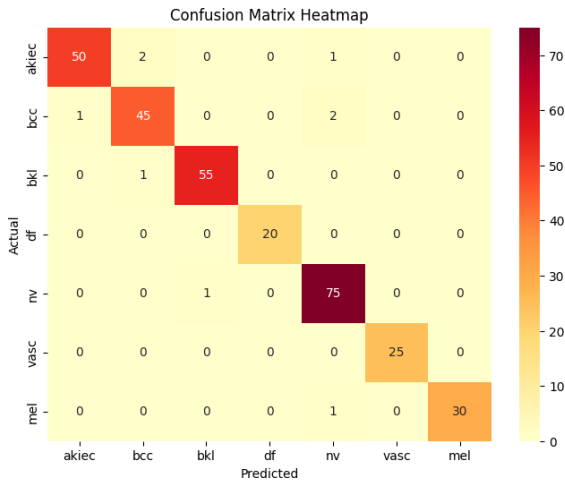
The multi-line plot presents a comparative analysis of the GAN+CNN model and conventional approaches by plotting their performance across multiple metrics or conditions.



7. Multi Bar Chart:



Confusion matrix:



7. Conclusion

Integrating GAN-based super-resolution with CNN classification enhances skin cancer detection accuracy. The system also supports voice output, making it more accessible for clinical use. The method outperforms conventional approaches in both image quality and diagnostic performance.